

SAT Hard Question 14

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Question: Solving Angle with Trig Identity

For two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$.

The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$ respectively.

What is the value of x ?

Answer Explanation: Trigonometric Identity

We are given that:

$$\cos(x + 61) = \sin(4x + 4)$$

Use identity:

$$\cos(\theta) = \sin(90^\circ - \theta) \Rightarrow \cos(x + 61) = \sin(90^\circ - (x + 61)) = \sin(29 - x)$$

So:

$$\sin(4x + 4) = \sin(29 - x) \Rightarrow 4x + 4 = 29 - x \Rightarrow 5x = 25 \Rightarrow x = 5$$

Question: Exponents and Roots

If a and b are numbers greater than 1 and $\sqrt[7]{a^6}$ is equivalent to $\sqrt{b^3}$, for what value of n is $a^{2n+1} = b$?

Answer Explanation: Rational Exponents

$$\sqrt[7]{a^6} = a^{6/7}, \quad \sqrt{b^3} = b^{3/2} \Rightarrow a^{6/7} = b^{3/2}$$

Raise both sides to $\frac{2}{3}$:

$$(a^{6/7})^{2/3} = b \Rightarrow a^{\frac{12}{21}} = a^{2n+1} \Rightarrow 2n+1 = \frac{12}{21} = \frac{4}{7} \Rightarrow n = \frac{4}{7} - \frac{1}{2} = \frac{-3}{14}$$

Question: Function and Proportional Constant

The function c gives the partial length, in centimeters (cm), of the circumference of a circle with a radius r cm:

$$c(r) = \frac{2}{5}(2\pi r)$$

For each increase of 18 cm in r , this partial length increases by $k\pi$ cm. What is the value of k ?

Answer Explanation: Rate of Change

Differentiate $c(r)$ with respect to r :

$$\frac{dc}{dr} = \frac{2}{5} \cdot 2\pi = \frac{4\pi}{5}$$

Then:

$$\text{Change in } c = \frac{4\pi}{5} \cdot 18 = \frac{72\pi}{5} \Rightarrow k\pi = \frac{72\pi}{5} \Rightarrow k = \frac{72}{5}$$

Question: Point on Graph of a Linear System

For each real number r , which of the following points lies on the graph of both equations in the xy -plane for the given system?

$$15x + 45y = 33$$

$$5x + 15y = 11$$

Options:

(A) $(\frac{11r}{5} - 3, r)$

(B) $(-3r + \frac{11}{5}, r)$

(C) $(r, -\frac{r}{3} - \frac{11}{15})$

(D) $(r, \frac{r}{3} + \frac{11}{15})$

Answer Explanation: Substitution in Linear System

Given the system:

$$15x + 45y = 33 \quad (1)$$

$$5x + 15y = 11 \quad (2)$$

Answer: B

Question: Solving Proportions for a Variable

If $\frac{a}{b} = \frac{4}{5}$ and $\frac{5b}{ta} = 35$, what is the value of t ?

Answer Explanation: Solve for t Using Substitution

We are given:

$$\frac{a}{b} = \frac{4}{5} \Rightarrow a = \frac{4}{5}b$$

Substitute into the second equation:

$$\frac{5b}{t \cdot a} = 35 \Rightarrow \frac{5b}{t \cdot \frac{4}{5}b} = 35 \Rightarrow \frac{5b}{\frac{4tb}{5}} = 35 \Rightarrow \frac{25}{4t} = 35 \Rightarrow 25 = 140t \Rightarrow t = \frac{25}{140} = \frac{5}{28}$$

$$t = \frac{5}{28}$$

Question: Unique Solution for Rational Equation

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

$$\frac{1}{2cx} = \frac{x}{16} + \frac{1}{c}$$

Answer

Answer: $c = -8$

Question: Consistency in Linear Systems

The given equation is one equation in a system of two linear equations. If the system has at least one solution, which of the following equations could be the other equation in the system?

$$7x + 5y = 2$$

Options:

- I. $10.5x + 7.5y = 3$
- II. $10.5x - 7.5y = 3$

Options:

- (A) I
- (B) II
- (C) I and II
- (D) Neither I nor II

Answer Explanation: Independent or Consistent Lines

Answer: C

Question: Exponential Function Point Match

The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ passes through the points $(c, 7)$ and $(2c, 117)$, where c is a constant. Which of the following could be the value of b ?

Answer Explanation: Using Given Points

Given: $f(x) = a^x - b$

From the point $(c, 7)$:

$$a^c - b = 7 \quad (1)$$

From the point $(2c, 117)$:

$$a^{2c} - b = 117 \quad (2)$$

Subtract (1) from (2):

$$a^{2c} - a^c = 110 \Rightarrow a^c(a^c - 1) = 110$$

Now try small integer values for a^c :

$$\text{Try } a^c = 11 \Rightarrow 11(11 - 1) = 110$$

So:

$$a^c = 11 \Rightarrow \text{From (1): } 11 - b = 7 \Rightarrow b = 4$$

$b = 4$

Question: Surface Area of a Rectangular Prism

A right rectangular prism has a base area of $16t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the prism is 9 cm.

What is the expression of the surface area, in cm^2 , of the right rectangular prism?

Answer Explanation: Surface Area Computation

Let width be w , then:

$$\text{Area of base} = lw = 16t \Rightarrow \frac{8}{3}w = 16t \Rightarrow w = \frac{16t \cdot 3}{8} = 6t$$

Now compute surface area:

$$SA = 2lw + 2lh + 2wh$$

Substitute values:

$$2lw = 2 \cdot 16t = 32t$$

$$2lh = 2 \cdot \frac{8}{3} \cdot 9 = 48$$

$$2wh = 2 \cdot 6t \cdot 9 = 108t$$

Add them:

$$SA = 32t + 48 + 108t = 140t + 48 \Rightarrow \boxed{140t + 48}$$

Question: Percentage Relationships of Masses

The mass of object A is 498% of the mass of object B, and the mass of object A is 0.830% of the mass of object C.

If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

Answer Explanation: Percent Equation Setup

Let:

- Mass of object B = x
- Then mass of object A = $4.98x$

Given: mass of A is 0.830% of mass of C:

$$4.98x = 0.0083 \cdot \text{mass of C} \Rightarrow \text{mass of C} = \frac{4.98x}{0.0083}$$

Now compute:

$$\text{mass of C} = 600x \Rightarrow \text{mass of C is } 600x \text{ and mass of B is } x$$

$$\Rightarrow p = \frac{600x}{x} \cdot 100\% = 60000\% \Rightarrow \boxed{p = 60000}$$

Question: Factoring a Quadratic Expression

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

Options:

- (A) $3x^2 + 7x + 14b$
- (B) $3x^2 + 16x + 14b$
- (C) $3x^2 + 18x + 14b$
- (D) $3x^2 + 25x + 14b$

Answer Explanation: Matching a Known Factor

We are told the expression has a factor of $x + 2b$.

Let the full factorization be:

$$3x^2 + mx + 14b = (x + 2b)(3x + r)$$

Use expansion:

$$(x + 2b)(3x + r) = 3x^2 + (3 \cdot 2b + r)x + 2br = 3x^2 + (6b + r)x + 2br$$

Now match with $3x^2 + mx + 14b$:

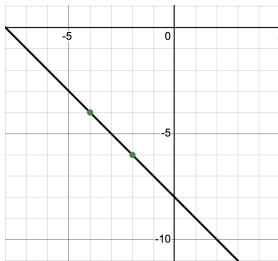
$$2br = 14b \Rightarrow r = 7$$

Then:

$$m = 6b + 7 \Rightarrow \text{Check values of } m \text{ in choices:}$$

Correct answer: D

Question: Interpreting a Linear Function



The graph of the linear function $y = f(x) - 19$ is shown.
If c and d are positive constants, which equation could define f ?

Options:

- (A) $f(x) = d - cx$
- (B) $f(x) = -d + cx$
- (C) $f(x) = d + cx$
- (D) $f(x) = -d - cx$

Answer Explanation: Line with Negative Slope and Vertical Shift

We are told that the graph shown is for $y = f(x) - 19$. That means the line shown is the function $f(x)$ shifted down by 19.

To recover $f(x)$, we shift the graph up by 19 units.

From the graph:

The line has negative slope $\Rightarrow f(x) = \text{something with } -cx$

We also observe that the y -intercept of the graph is approximately -4 (which becomes 15 after adding 19).

So the general form must be:

$$f(x) = d - cx, \quad \text{with } d > 0, c > 0$$

Hence, the correct form is:

$$\boxed{f(x) = d - cx} \Rightarrow \textbf{Answer: (A)}$$

Question: Percent Decrease as a Factor

The expression kx represents the result of decreasing a positive force, x , acting on an object by 0.34%.

What is the value of k ?

Options:

- (A) 0.34%
- (B) 1.34%
- (C) 99.66%
- (D) 100.34%

Answer Explanation: Subtracting Percent from 100%

The expression kx reflects a reduction of x by 0.34%.
That means k must represent $100\% - 0.34\% = 99.66\%$ of x :

$$k = 1 - 0.0034 = 0.9966 \Rightarrow \boxed{k = 99.66\%}$$

Answer: (C)

Question: Interior Angle Expression in Terms of p

A polygon has exactly 87 sides. If the measure of each of the 87 interior angles of this polygon is $(180p)^\circ$, what is the value of p ?

Options:

- (A) $\frac{85}{87}$
- (B) $\frac{37}{87}$
- (C) $\frac{87}{90}$
- (D) $\frac{87}{180}$

Answer Explanation: Interior Angles of a Polygon

For an n -gon, the sum of interior angles is:

$$(n - 2) \cdot 180^\circ$$

If each angle is equal:

$$\text{Each angle} = \frac{(n - 2) \cdot 180}{n}$$

Given $n = 87$, so:

$$\text{Each angle} = \frac{(87 - 2) \cdot 180}{87} = \frac{85 \cdot 180}{87}$$

Also given:

$$\text{Each angle} = 180p \Rightarrow 180p = \frac{85 \cdot 180}{87} \Rightarrow p = \frac{85}{87}$$

$$\boxed{p = \frac{85}{87}}$$

(Answer A)

Question: Exponential Sequence Term Comparison

The function $f(n) = 7(20.44)^{\frac{n}{4}}$ gives each term of a sequence as a function of the term's position n in the sequence, where n is a whole number.

The value of the term in position 14 is $p\%$ more than the value of the term in position 10.

What is the value of p ?

Answer Explanation: Percent Increase Between Terms

We are given:

$$f(n) = 7(20.44)^{\frac{n}{4}}$$

Find:

$$f(10) = 7(20.44)^{\frac{10}{4}} = 7(20.44)^{2.5}$$

$$f(14) = 7(20.44)^{\frac{14}{4}} = 7(20.44)^{3.5}$$

Then compute:

$$\frac{f(14)}{f(10)} = \frac{7(20.44)^{3.5}}{7(20.44)^{2.5}} = (20.44)^1 = 20.44$$

So:

$f(14)$ is 20.44 times $f(10) \Rightarrow$ increase is $20.44 - 1 = 19.44 \Rightarrow p = 1944\%$

$p = 1944$

Question: Determining the Height of a Pyramid

A right square pyramid has a total surface area of 28,160 square inches, and the combined surface area of the four lateral faces of this pyramid is 16,060 square inches.

What is the height, in inches, of this pyramid?

Options:

- (A) 48
- (B) 55
- (C) 73
- (D) 110

Answer Explanation: Solve with Geometry Formulas

We are told:

- Total surface area: 28,160
- Lateral surface area: 16,060

The base area is:

$$\text{Base area} = 28,160 - 16,060 = 12,100$$

$$\Rightarrow \text{Base is a square, so: } s^2 = 12,100 \Rightarrow s = \sqrt{12,100} = 110$$

Explanation

Each lateral face is a triangle:

$$\text{Area of one face} = \frac{1}{2} \cdot 110 \cdot \ell, \text{ where } \ell \text{ is slant height}$$

$$\Rightarrow 4 \cdot \frac{1}{2} \cdot 110 \cdot \ell = 16,060 \Rightarrow 220\ell = 16,060 \Rightarrow \ell = 73$$

Now use Pythagoras to find height h :

$$h^2 + \left(\frac{110}{2}\right)^2 = 73^2 \Rightarrow h^2 + 55^2 = 5329$$

$$\Rightarrow h^2 + 3025 = 5329$$

$$\Rightarrow h^2 = 2304$$

$$\Rightarrow h = \sqrt{2304} = \boxed{48}$$

Question: One Real Solution Condition for Radical Equation

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

$$\sqrt{k - x} = 32 - x$$

Answer Explanation: Graphical and Algebraic Intersection

We are given:

$$\sqrt{k-x} = 32-x$$

****Step 1:**** Domain conditions:

$$k-x \geq 0 \Rightarrow x \leq k \quad \text{and} \quad 32-x \geq 0 \Rightarrow x \leq 32$$

So the solution must satisfy $x \leq \min(k, 32)$.

****Step 2:**** To have ****exactly one solution****, the graphs of both sides must be tangent at that solution.

Square both sides:

$$k-x = (32-x)^2 = 1024 - 64x + x^2$$

Bring all terms to one side:

$$x^2 - 63x + 1024 - k = 0$$

For exactly one solution, discriminant must be zero:

$$\Delta = (-63)^2 - 4(1)(1024 - k) = 3969 - 4096 + 4k = 0 \Rightarrow 4k = 127 \Rightarrow \boxed{4k = 127}$$